

PUNITHRAJ

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SUMMARY

AI Engineer with 2+ years building production LLM, RAG, and agentic AI systems for voice and conversational hiring platforms. Skilled in Python, LangGraph, FastAPI, and cloud-native deployment, with a consistent record of turning AI systems into measurable business outcomes — from 3x productivity gains to 75% cost reductions.

EXPERIENCE

AI Engineer — TCS, Bengaluru

Feb 2026 – Present

- **Unified Voice + WhatsApp Hiring Pipeline:** Connected the voice bot and WhatsApp chatbot into one continuous hiring flow with cross-channel conversation memory and a Redis-based nudge engine that escalates text to voice callbacks over 24 hours — lifting connect-to-interest conversion by 40%, WhatsApp response rate from 21% to 30%, and hiring productivity by 3x.
- **AI Voice Recruiter Bot (Outbound & Inbound):** Built and scaled an AI recruiter calling bot for blue-collar hiring and intent verification using Pipecat across two telephony providers, achieving <1.2s average bot latency in production.

Associate AI Engineer — Cognizant, Bengaluru

Jan 2025 – Jan 2026

- **Performance & Scalability:** Re-architected the call lifecycle into async workflows with a queue-based predictive dialer handling 60K+ daily calls, cutting p50 call duration from 4.5h to 1.5h, and introduced a centralized Milvus-based TTS cache that cut TTS costs by 75% and audio lag by 0.7s.
- **WhatsApp Recruiter Chatbot (0→1):** Built a LangGraph-based hiring chatbot with 20+ flow variants — intent check, doubt resolution, onboarding, FAQ — handling voice notes, multi-language responses, and template-based outreach via WhatsApp Business APIs.
- **Growth, Experiments & Analytics:** Drove a 35% improvement in goal completion through A/B testing of bot personalities and script logic, using ClickHouse for conversational analytics, n8n for workflow automation, and NotebookLM for LLM-assisted analysis.
- **Multi-Agent Healthcare Chatbot:** Built a multi-agent RAG chatbot handling medical, product, and sales queries with long-term user context, deployed via event-driven workflows with cloud functions, scalable storage, and CI/CD pipelines on GCP.
- **Report Explanation via Video:** Engineered an automated video health-report explainer using FFmpeg, ElevenLabs, and Python Lambda functions, improving report comprehension and customer engagement.

Junior AI Engineer — Tech Mahindra, Bengaluru

Nov 2023 – Dec 2024

- **Sales Intelligence Platform:** Built AI-powered sales call audit and assist modules analyzing call transcripts and customer data, integrating insights into ERP systems — generating 23% extra revenue while cutting customer escalations by 47%.
- **Business Process Automation & Email Agent:** Developed an AI-driven customer support email agent achieving 60%+ first-response resolution and automated internal analytics workflows using similarity-based matching, cutting manual effort from 8 hours to 20 minutes per cycle.
- **Microservices Migration:** Migrated a monolithic PHP system to Golang (Gin/GORM), using goroutines to reduce average API latency from 600ms to <200ms.
- **Marketing Automation:** Automated large-scale WhatsApp/SMS/Email communication via PHP cron jobs and optimized Redis/MySQL query performance for high-traffic segments.

TECHNOLOGIES

AI & ML: LLMs, RAG, Generative AI, Prompt Engineering, Machine Learning, Deep Learning, Neural Networks, CNN, NLP, Computer Vision

Frameworks: LangGraph, LangChain, FastAPI, PyTorch, TensorFlow, Scikit-Learn, Streamlit, Pandas, NumPy

Languages: Python, SQL

Tools & MLOps: Docker, Git, GitHub, Model Deployment, Vector Databases, REST APIs, Data Preprocessing, Data Analysis

EDUCATION

Bachelor of Computer Applications (BCA)

June 2020 – June 2023

Mahatma Gandhi University, Kottayam — CGPA: 6.87